Compressive Sensing and Image Denoising

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Problem Reformulation

From measurements to an ill-posed inverse problem

- We want to recover an unknown image $x \in \mathbb{R}^n$ from partial observations $y \in \mathbb{R}^m$ $(m \ll n)$.
- Forward model (sub-sampling): $y = Ax$, where A selects a subset of pixels.
- Since $m \ll n$, the problem is **underdetermined** $\Rightarrow$ many solutions satisfy $Ax = y$.
- **Goal of this lab**: An image can be reconstructed from very few measurements if it has a sparse representation.

Key prior and Optimization problem

- Let $\mathcal{F}$ be the 2D Fourier transform and define: $\mathcal{X} = \mathcal{F}x$.
- Ideal formulation:

$$\min_{\mathcal{X}} \|\mathcal{X}\|_0 \quad s.t. \quad A\mathcal{F}^{-1}\mathcal{X} = y \tag{1}$$

- Convex relaxation:

$$\min_{\mathcal{X}} \|\mathcal{X}\|_1 \quad s.t. \quad A\mathcal{F}^{-1}\mathcal{X} = y \tag{2}$$

Baseline

Greedy Method

- Iterative algorithm for solving compressed sensing problems.
- Main idea:
  - Start with empty solution (all zeros).
  - At each iteration: select the "most important" Fourier coefficient (highest correlation with residual).
  - Update solution and continue until convergence.
- Advantages: Simple and easy to implement.
- Disadvantages: No optimality guarantee and Computation cost is high!

Greedy Algorithm

- Initialize: $X_0 = 0, \qquad S_0 = \varnothing$
- Compute residual: $r_k = y - \Phi X_k$
- Compute proxy: $c_k = \Phi^H r_k$
- Select one dominant entry and update support: $i_k = \arg\max_i \left| c_i^{(k)} \right|, \qquad S_{k+1} = S_k \cup \{i_k\}$
- Refit coefficients on the support (least squares): $X_{S_{k+1}}^{k+1} = \arg\min_u \left\| y - \Phi_{S_{k+1}} u \right\|_2$

Douglas-Rachford Algorithm

- Reformulation for Douglas-Rachford splitting

$$\mathcal{X}^\star \in \arg\min_{\mathcal{X}\in\mathbb{C}^n} \left(f_1(\mathcal{X}) + f_2(\mathcal{X})\right) = \arg\min_{\mathcal{X}\in\mathbb{C}^n} \left(\|\mathcal{X}\|_1 + \iota_{\{y=\mathcal{A}\mathcal{X}\}}(\mathcal{X})\right) \tag{3}$$

- For the problem at hand, we evaluate the proximal operators of

$$\begin{cases} \gamma f_1(\mathcal{X}) \equiv \gamma\|\mathcal{X}\|_1, \\ \gamma f_2(\mathcal{X}) \equiv \iota_{\{y=\mathcal{A}\mathcal{X}\}}(\mathcal{X}) \end{cases} \Longrightarrow \begin{cases} \text{prox}_{\gamma\|\cdot\|_1}(\mathcal{Y})_{ij} = \begin{cases} \left(1 - \dfrac{\gamma}{|\mathcal{Y}_{ij}|}\right)\mathcal{Y}_{ij} & |\mathcal{Y}_{ij}| > \gamma \\ 0 & |\mathcal{Y}_{ij}| \leq \gamma \end{cases} \\ \text{prox}_{\gamma f_2}(\mathcal{X}) = \mathcal{X} + \mathcal{A}^\top(\mathcal{A}\mathcal{A}^\top)^{-1}\left(y - \mathcal{A}\mathcal{X}\right) \end{cases} \tag{4}$$

- **We have 2 parameters**: step/prox parameter $\gamma > 0$ and relaxation $\rho \in (0, 1)$.
- **Algorithm**: with the iterative variable $\mathcal{U}^k$:
  - Proximal step for f_1 (soft-thresholding): $\mathcal{X}^k = \text{prox}_{\gamma f_1}\left(\mathcal{U}^k\right)$, phase is preserved.
  - Reflection followed by the proximal step for f_2 (projection): $\mathcal{Z}^k = \text{prox}_{\gamma f_2}\left(2\mathcal{X}^k - \mathcal{U}^k\right)$.
  - Update: $\mathcal{U}^{k+1} = \mathcal{U}^k + \rho\left(\mathcal{Z}^k - \mathcal{X}^k\right)$.
- Recovered image: $x_k = \mathcal{F}^{-1}\mathcal{X}_k$.

Comparision of Greedy method and Douglas-Rachford Method

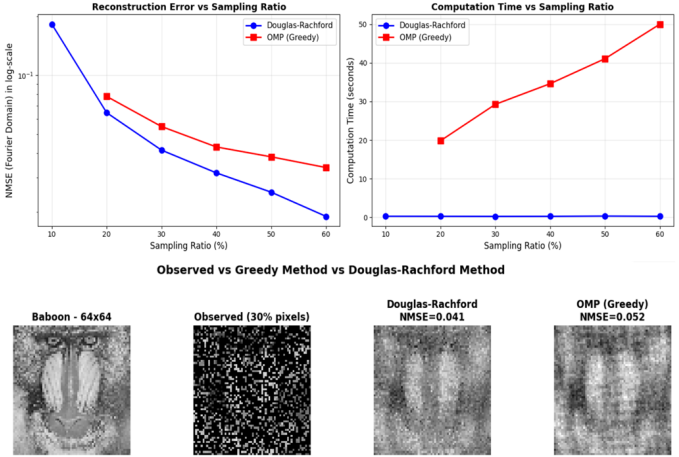


Figure: NMSE vs Sampling Ratio (same masks/seed) with Image shape = 64x64, $\gamma = 0.5$, $\rho = 1.0$, K = 0.05

Fine tuning Douglas-Rachford Method

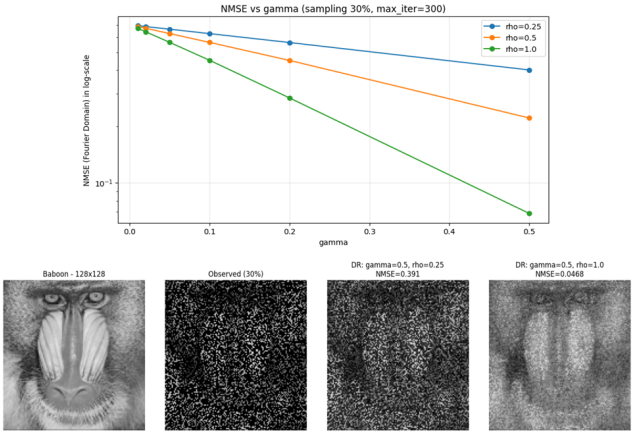


Figure: Effect of the free parameter in visual image

Conclusion

- Problem: recover an image from few measurements by exploiting Fourier sparsity and solving the convex problem.
- Methods:
  - Greedy baseline: no optimality guarantee and computation cost is very high.
  - Douglas-Rachford: proximal splitting using *soft-thresholding* L_1 and *projection* on to the measurement constraint.
- Results:
  - Higher sampling ratio improves reconstruction.
  - Douglas-Rachford method often yields more coherent visual recovery while greedy method can be competitive in NMSE depeding on settings.
- Free parameters: γ controls threshold strength while $\rho \in (0, 1)$ controls convergence speed. Tuning these parameters impacts both NMSE and visual quality.

Thank you for your attention!